Laboratory Protocols, to reproduce the Results

In-silico Lab Protocols:

Pharmacophore Modeling :

Structure-based individual Pharmacophore mapping from PDB 3LPF:

Two and three dimensional (2D and 3D) structure-based pharmacophore models were derived from available PDB I.D 3LPF using software Ligand Scout 3.0 version.

Following steps were included,

Extraction of PDB I.d (3LPF) by Ligand Scout

Double Click on the highlighted yellow box

Click on the option "Generate the pharmacophore feature, from the upper bar of software

The software generated the 2D and 3D structure-based Pharmacophore model in middle and right pane along with the highlight Pharmacophore features, (Fig 6a-6b).

Structure-based individual Pharmacophore mapping from PDB 3LPG:

2D and 3D structure-based pharmacophore models were derived from available PDB 3LPG using Ligand Scout software 3.0 version, Following Steps were included.

Extraction of the PDB I.D (3LPG)

Double Click on the highlighted yellow box

Click on the option "Generate the pharmacophore feature, from the upper bar of software

The software generated the 2D and 3D structure-based Pharmacophore models along with the highlight Pharmacophore features, (Fig 7a-7b).

Structure-based individual pharmacophore mapping of 3K4D:

2D and 3D structure-based pharmacophore models were derived from 3K4D using Ligand Scout 3.0 version.

Extraction of the PDB I.D (3K4D)

Double Click on the highlighted yellow box

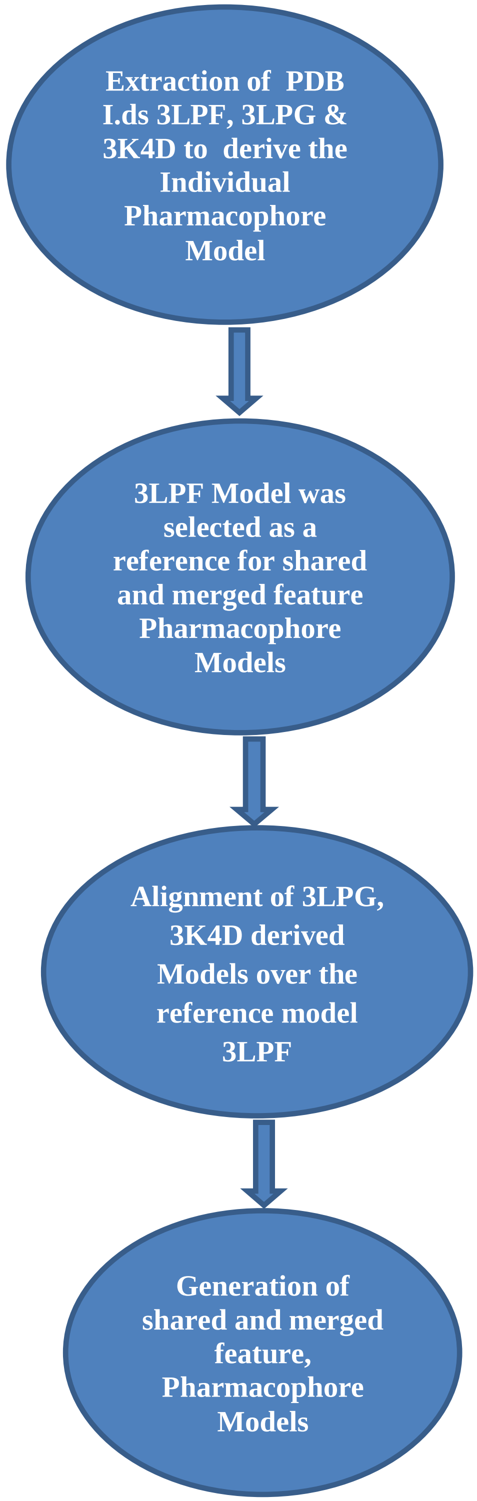
Click on the option "Generate the pharmacophore feature, from the upper bar of software

The software generated the 2D and 3D structure-based Pharmacophore models along with the highlight Pharmacophore features (Fig 8a-8b).

Generation of structure-based shared and merged feature pharmacophore models:

A five-element structure-based shared feature pharmacophore model was also derived with the help of all three individual derived models. For this purpose, we selected the derived model of 3LPF as reference, due to its potent inhibitor bound with IC_{50} value of 326 nM to extract the keen chemical features of structure-based pharmacophore model by setting it as a reference. The remaining two, individual derived models 3LPG and 3K4D, were aligned over it, and finally a shared featured structure-based five-element Pharmacophore model was generated along with the descriptive features. Similarly, a merged feature model was also generated by extracting all the pharmacophore features together [Fig 9a-9b].

General Pharmacophore generation scheme by using software Ligand Scout Version 3.0:



Pharmacophore-based Virtual screening:

In the present study, five structure-based pharmacophore models were derived with the help of three available E.coli β-glucuronidase PDBs. 3LPF, 3LPG and 3K4D by using Ligand Scout software 3.0 [22]**.** The models were used, to search chemical *data-base* of ICCBS by using software Molecular Operating Environment (MOE) (2010-212). MOE is the most useable software in computational based drug designing and discovery process designed by Chemical Computing Group. The software is successfully using in structure-based drug design, fragment-based drug design, Pharmacophore mapping, protein and anti-body modeling and 3DQSAR (three dimensional quantitative structure activity relationship). The *Data-base* was at first filtered to obtain *drug-like* candidates according to Lipiniski ROF by using MOE. Filter removed all of those compounds which deviate from ROF and does not follow drugability criteria. After filtration of *in-house*, *data-base* 8,262 *drug-like* filtered candidates were obtained which further used for pharmacophore-based virtual screening. All of the derived models were saved into (.Ph4) MOE compatible format. The *data-base* was saved into (.mdb) format. The software retrieved the Pharmacophore models and searched the query editor to scaffold hopping of chemical database. Pharmacophore-based virtual screening identified altogether 1,249 hits [Figure 2].

Molecular docking Protocol:

Selection and preparation of receptor protein:

Selection of protein (*β*-glucuronidase) particular structure for docking was done on the basis of protein-ligand complex resolution of the X-ray co-crystallize structure and redocking experiment. Protein PDB I.D 3LPF was used to perform redocking with Z77 (thiourea derivative) bound ligand of resolution 2.26 Å. It is a homo dimmer protein which consists of two chains, chain A and chain B with similar catalytic amino acid residues sequence. We selected chain A with conserved water molecules HOH, 655 and 733 respectively [23].

Catalytic residues:

Catalytic traid amino acid residues include two acidic amino acids glutamic acid Glu540 and Glu451 and a tyrosine residue Tyr504. Apart from catalytic traid residues, some other interactive amino acid residues have also been reported which non-covalently interacts with ligand such as, amino acid Glu413A, Leu361, Phe448A and Tyr472 [24].

Reported inhibitors *data-set*:

We selected 66 already reported known inhibitors small *data-set* from the literature of *β*-glucuronidase belongs to different classes of compounds (list provided in supporting information). All of the compounds 2D (.mol) format structures were converted into 3D (.mol_2) format by using open babel *command-line* on OS Linux. Minimization of compounds was performed by using SYBYL (Tripose) to generate low energy stable conformers.

Molecular Docking:

Molecular Docking studies of Pharmacophore based virtually screened 1,249 hits along with 66 reported inhibitors against targeted receptor *β*-glucuronidase was performed by using FRED software, *FRED* uses multi-conformer docking algorithm, which generates a set of low-energy conformers separately, and then do rigid docking of each conformer. *FRED*, is a non-stochastic software and finds the most favorable ligand pose *via* a systematic exploration of conformational and rotational space Chemgauss-4 is an optimized scoring function of *FRED* Chemgauss-3, The interactions which can be scored by Chemgauss-4 scoring functions includes steric, H-acceptor, H-donors, coordinating groups, metals, lone pairs, polar hydrogens, chelator coordinating groups and overall total H-bonding score between ligand and protein, higher the magnitude of Chemgauss-4 represents the more tightly binding of ligand with receptor through non-covalent forces of interactions,[25]. The *data-set* was subjected for docking by using FRED software, which successfully docked 1,315. Molecules after completion of docking.

Enrichment factor: "*Enrichments smaller than 1 are indicative of a preferred selection of inactive compounds, corresponding to an enrichment of active compounds at the bottom of the sorted library*" [26].

E.F was calculated for 5%, 10%, 15% and 20% of docked molecules by using the following formula, [Figure 3a-3d].

$$\text{Enrichment factor of a (subset)} = \frac{Hit_s}{N_s} \Big/ \frac{Hit_t}{N_t}$$

Or

E.F= no of actives in % of *data-base* /% of *data-base complete *data-base*/total no of known actives**

Where Hit_s = no of actives in a subset, N_s = total no of compounds in a subset, Hit_t = total no of actives in the *data-base*, N_t = total no of compounds in the data base.

By using the above formula, we calculated the enrichment factor for 5%, 10%, 15% and 20 % for each scoring function of structure-based hits from *in-house data-base* [Table 1].

Rescoring by using genetic optimization for ligand docking (*GOLD*) Software:

As the new version of *FRED* consists of only a single scoring function Chemgauss-4. It is the modified form of Chemgauss-3 scoring function. Therefore, to get more information rescoring of *FRED* scoring function Chemgauss-4 score was performed by using *GOLD* software 5.1 version [27], into Gold-score, Chem-score and ASP (Astex Statistical Potential) scoring functions. Gold scoring functions are dimensionless. However, in each case, the magnitude of score indicates how best the ligand bound with receptor, higher the value of score represents the best binding of ligand with receptor. The Gold score fitness function is the original scoring function of *GOLD*. It deals with ligand binding positions and non-covalent binding interactions such as H-bonding energy, van der waals energy, metal interaction and ligand torsion strain. While, the Chem-score fitness function deals with dG that represents total free energy change. It also incorporates a protein-ligand atom clash term and an internal energy term. Chem-score takes account of hydrophobic-hydrophobic contact area, hydrogen bonding, ligand flexibility and metal chelating interactions. After rescoring comparative enrichment factors were calculated for 5%, 10%, 15% and 20% of each scoring function. It was observed that out of these four scoring functions, Chemgauss-4 was found to be as a best within the 5% enrichment of data-base, while for the rest of 10%, 15% and 20% scoring function, Chem-score was found to be as dominant one, comparative to GOLD score and ASP score.(Astex Statistical Potential) [Table 1].

In-Vitro Protocols:

***β*-Glucuronidase inhibition assay protocol:**

Top ranked (5% enrichment) (out of 68 compounds, 33 were subjected) for β-Glucuronidase inhibitory activity was determined with the help of spectrophotometric method by measuring the absorbance at 405 nm of *p*-nitrophenol formed from the substrate (*p*-nitrophenyl-*β*-D-glucuronide, N1627-250 mg, Sigma Aldrich). The total reaction volume was 250 µL. The compound dissolved in DMSO (100%), which becomes 2% in the ultimate assay (250 µL) and the similar conditions were used for standard (D-saccharic acid 1, 4-lactone, Sigma Aldrich). The reaction mixture contained 185 µL of 0.1 M acetate buffer, 5 µL of test compound solution, 10 µL of (1U) enzyme solution (G7396-25KU, Sigma Aldrich) was incubated at 37°C for 30 min. The plates were read on a multiplate reader (SpectraMax plus 384) at 405 nm after the addition of 50 µL of 0.4 mM *p*-nitrophenyl-*β*-D-glucuronide. All assays were performed in triplicate. IC_{50} values were calculated by using EZ-Fit software (Perrella Scientific Inc., Amherst, MA, U.S.A.). These values are the mean of three independent readings [28] [Table 3].

Cytotoxicity assay Protocol:

Cytotoxic activities of eleven potent inhibitors were evaluated in 96-well flat-bottomed micro plates by using the standard MTT (3-[4, 5-dimethylthiazole-2-yl]-2, 5-diphenyl-tetrazolium bromide) colorimetric assay [29]. For this purpose, 3T3 (mouse fibroblast) cells were cultured in Dulbecco's Modified Eagle Medium, supplemented with 5% of fetal bovine serum (FBS), 100 IU / ml of penicillin and 100 µg / ml of streptomycin in 75 cm^2 flasks, and kept in 5% CO_2 incubators at 37^oC. Exponentially growing cells was harvested, counted with haemocytometer and diluted with a particular medium. Cell culture with the concentration of $5x10^4$cells / ml was prepared and introduced (100 µL / well) into 96-well plates. After overnight incubation, medium was removed and 200 µL of fresh medium was added in different concentrations of compounds (1-30µM). After 48 hrs, 200 µL MTT (0.5 mg / ml) was added to each well plate and incubated further for 4 hrs. Subsequently, 100µL of DMSO was added to each well. The extent of MTT reduction to formazan within cells was calculated by measuring the absorbance at 540 nm, using a micro plate reader (Spectra Max plus, Molecular Devices, CA, USA). The cytotoxicity was recorded as concentration causing 50% growth inhibition (IC_{50}) for 3T3 cells. The percent inhibition was calculated by using the following formula.

% inhibition = 100-((mean of O.D of test compound – mean of O.D of negative control) / (mean of O.D of positive control – mean of O.D of negative control)*100).

The results (% inhibition) were processed by using Soft- Max Pro software (Molecular Device, USA) [Table 4].